

What is Web Application Security

- It means protecting web applications from attacks that steal or damage data.
- Web security ensures that apps behave as intended, even when attackers try to misuse them.
- It involves finding and fixing vulnerabilities in code, servers, and configurations.



Why Web Security is Important?

-  Millions of users depend on web apps for banking, shopping, healthcare, etc.
-  One weak spot can allow hackers to steal sensitive data or take control of systems.
-  Security failures can cause financial loss, legal issues, and brand damage.
-  A secure web app builds user confidence and protects business reputation.

How Web Applications Work?

- A client (browser) sends a request to the server using HTTP/HTTPS.
- The server processes that request (maybe connects to a database).
- It then sends back a response, usually HTML, JSON, or other data.
-  This back-and-forth flow is where attackers try to inject malicious data (like SQL Injection or XSS).

Common Web Application Threats

-  SQL Injection (SQLi) – attacker manipulates database queries.
-  Cross-Site Scripting (XSS) – injecting malicious scripts into web pages.
-  Cross-Site Request Forgery (CSRF) – forcing users to perform unwanted actions.
-  Broken Authentication – stealing or guessing login details.
-  Insecure File Uploads – uploading dangerous files to servers.
-  Path Traversal – accessing restricted files or directories.

Security Mindset & Developer Responsibility

- Security is not just for hackers, it's every developer's responsibility.
- Think like an attacker to write safer code.
- Always validate input, sanitize output, and follow secure coding standards.

Mindset Difference

-  A normal developer thinks:
“How can I make it work?”
-  A security-focused developer thinks:
“How can someone break this?”
-  True web security starts when developers think like attackers and code like defenders.